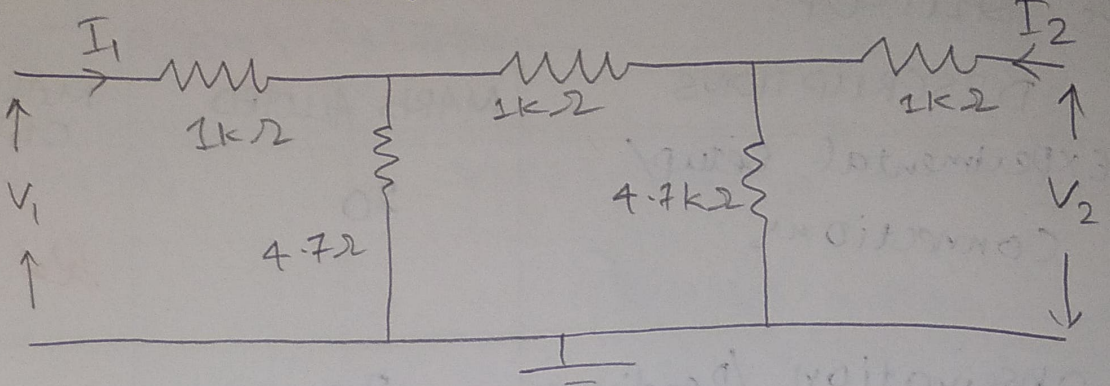
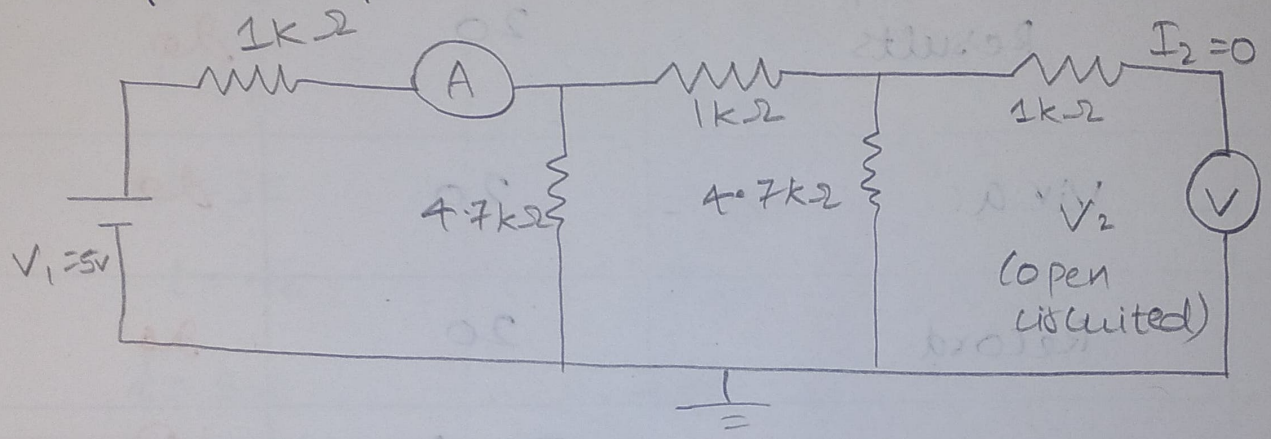
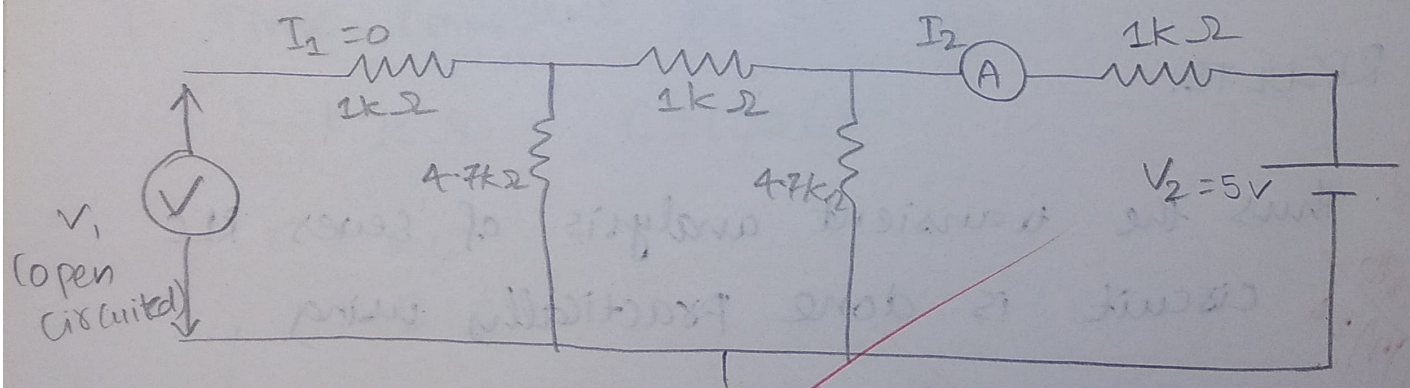
CIRCUIT DIAGRAM:



Output open circuited:



Input open circuit:



EXP. NO: 11 DATE: 11.03.2026	DETERMINATION OF Z PARAMETERS OF TWO PORT NETWORK

AIM:

To calculate and verify Z parameters of a two port network using Multisim Simulator.

APPARATUS REQUIRED:

S.NO	COMPONENTS	SPECIFICATION	QUANTITY
1	Power supply	-	1
2	Multimeter	-	1
3	Resistors	10kΩ, 10kΩ, 4.7Ω	
4	Bread board	-	1
5	Connecting wires PC with Multisim Simulator	-	-

S.NO	Output open circuited ($I_2=0$)			$Z_{21} = V_2/I_1$ (Ω)	$Z_{11} = V_1/I_1$ (Ω)
	V_1	V_2	I_1 (mA)		
1	5.0	2.2	1.0	~~500~~ 2200	5000

S.NO	Input open circuit ($I_1 = 0$)			$Z_{12} = V_1/I_2$ (Ω)	$Z_{22} = V_2/I_2$ (Ω)
	V_2	V_1	I_2		
1	5.0	2.1	1.0	2100	5000

OB verified

PROCEDURE:

1. Connect the circuit as shown in fig. & switch 'ON' the experimental Board.
2. First open the O/P terminal & supply 5V to I/P terminal. Measure O/P voltage & I/P current
3. Secondly, open I/P terminal & supply 5V to O/P terminal. Measure I/P voltage & O/P current using multi-meter.
4. Calculate the values of Z parameter using Eq (1) & (2)
5. Switch "OFF" the supply after taking the readings.

MARKS SPLIT-UP:

SI NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1.	Experimental setup/connections	20	20
2.	Observation/ Readings	20	20
3.	Calculations/ Results.	20	20
4.	Viva	20	20
5.	Record.	20	20
		Total	100

Theoretical verification:

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

when $I_2 = 0$: $Z_{11} = \frac{V_1}{I_1}$, $Z_{21} = \frac{V_2}{I_1}$

$$Z_{11} = \frac{5}{1mA} = 5k\Omega$$

$$Z_{21} = \frac{2.2}{1mA} = 2.2k\Omega$$

when $I_1 = 0$: $Z_{22} = \frac{V_2}{I_2}$, $Z_{12} = \frac{V_1}{I_2}$

$$Z_{22} = \frac{5}{1mA} = 5k\Omega$$

$$Z_{12} = \frac{2.1}{1mA} = 2.1k\Omega$$

10

11/3/26

RESULT:

The z-parameters of the two ~~parameter~~ port network has been calculated and verified by using Multisim Simulator.